WW-C-626b
October 21, 1966

SUPERSEDING
Int. Fed. Spec. WW-C-00626a (GSA-FSS)
May 14, 1964 and
Fed. Spec. WW-C-626
February 7, 1933

FEDERAL SPECIFICATION

COUPLING, HOSE (OIL-SUCTION AND DISCHARGE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers four types of hose couplings for oil-suction and discharge.

1.2 Classification. Coupling shall be of the following types and sizes, as specified (see 6.2):

1.2.1 *Type:*
Type I—Slotted expansion.
Type II—Lug, ribbed shank.
Type III—Iron pipe nipples.
Type IV—Iron pipe (banded).

1.2.2 *Sizes.*
Type I—4 inches only.
Type II—1½, 2, 2½, 3, 3½, and 4 inches.
Types III and IV—2, 2½, 3, 3½, 4, 5, 6, 8, 10, and 12 inches.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-B-621—Brass Castings, Leaded Yellow.
QQ-B-626—Brass, Leaded and Non-leaded: Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strip).
WW-P-406—Pipe, Steel (Seamless and Welded) (For Ordinary Use).
WW-P-441—Pipe; Wrought Iron, (Welded, Black or Zinc-Coated).
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-T-76—Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.
MIL-B-121—Barrier Material, Greaseproofed, Waterproofed, Flexible.

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Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H28—Screw-Thread Standards for Federal Services, Part II.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.)

3. REQUIREMENTS

3.1 Design. Couplings shall conform to the respective figures and tables incorporated herein. Threads shall conform to sections VII and X of NBS H-28, part II.

3.2 Finish. All nonmating surfaces shall be finished reasonably smooth by grinding, polishing, shot blasting, or sandblasting, and shall be free from burrs. All mating and gasket surfaces shall be 125 r.m.s. finish or better and all machine tolerances, plus or minus 0.010 inch. The corrugations of both male and female sections of coupling shall be left rough so as to bind the hose.

3.3 Parts. A complete set of types I, II, III, and IV couplings shall consist of the following parts:

Type I:
Part 1. Female section.
Part 2. Male section.
Part 3. Swivel of female section.
Part 4. Expansion rings, one each for male and female sections.
Part 5. Rubber washer, fitting recess in swivel.

Type II:
Part 1. Female section.
Part 2. Male section.
Part 3. Swivel of female section.
Part 4. Clamps for attaching hose.
Part 5. Washer, fitting recess in swivel.

Type III:
Part 1. Nipple.
Part 2. Clamp, if specified (see 6.2).
Part 3. Standard companion flange, if specified (see 6.2).

Type IV:
Part 1. Nipple.
Part 2. Standard companion flange, if specified (see 6.2).

3.4 Materials.

3.4.1 *Types I and II.* All metal parts except expansion rings shall be drop-forged brass conforming to QQ-B-626, composition 21, cast yellow brass conforming to composition C of QQ-B-621, or of a yellow brass conforming to chemical and physical properties in tables I and II.

TABLE I. *Chemical composition*

	Minimum	Maximum
Copper, percent	82.00	84.00
Tin, percent	2.5	3.5
Lead, percent	2.5	3.5
Zinc, percent	9.0	11.0
Phosphorus, percent		0.03
Antimony, percent		0.25
Nickel, percent	0.25	0.50
Other elements, other than copper, tin, lead, zinc, and nickel, percent[1]		0.50

[1] Includes iron maximum percent 0.35.

TABLE II. *Physical properties*

	Minimum	Maximum
Tensile strength, lb./in.	31,000	
Elongation in 2 inch, percent	25	
Reduction in area, percent	20	30
Brinell hardness (500 kg. for 30 seconds)	50	60

3.4.2 *Type I, part 4 expansion rings.* Expansion rings used in type I coupling shall

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be made from seamless drawn brass tubing, conforming to either of the compositions in Table III.

TABLE III. *Material composition*

Copper	Tin (max.)	Zinc	Lead (max.)	Iron (max.)
Percent	*Percent*	*Percent*	*Percent*	*Percent*
79 to 86		Remainder	0.20	0.05
59 to 68	1.5	Remainder	3.50	.20

3.4.3 *Types III and IV.* Nipples shall be of steel pipe conforming to WW-P-406, weight A, or wrought iron pipe conforming to WW-P-441, class A. Bands required for type IV coupling shall be of steel and welded all around the diameter of coupling. Clamps, when required, shall be of brass having a copper content of not less than 68 percent copper.

3.5 Hydrostatic pressure. The couplings shall be capable of withstanding the respective hydrostatic pressure for the sizes listed following:

Size	Pressure
Inches	*Pounds*
4	250
6	200
8	150

The pressures for other sizes shall be the same as required for the hose.

3.6 Workmanship. Couplings shall be free of burrs, cracks, sharp edges, pits or any other defects which would affect their serviceability for their intended purpose.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. The sample size levels of inspection shall be in accordance with MIL-STD-105.

4.2.1 *Lot.* Unless otherwise specified (see 6.2), a lot shall consist of all couplings of the same size and type submitted for inspection at the same time.

4.2.2. *Visual and dimensional examination of end item.* Sampling of couplings for these characteristics shall be in accordance with inspection level S-4.

4.2.3 *Samples for test.* Sampling of coupling for test shall be in accordance with inspection level S-2.

4.3 Inspection.

4.3.1 *Examination of end item.* The samples selected in 4.2.2 shall be examined for defects as listed in table IV. The acceptable quality levels (AQL's) shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

TABLE IV. *Classification of defects*

Defects	Major	Minor
Not the type or size specified	X	
Component parts missing	X	
Dimension not in accordance with applicable dimensions and figures for type and size specified	X	
Cracks in couplings	X	
Metal protruding on parting or flash line of cast or forged units	X	
Threads rough and eccentric	X	
Sharp burrs or slivers	X	
Porosity visible on surface		X
Assembly damaged or defective not affecting function or serviceability		X

4.3.1.1 *Inspection of preparation for delivery requirements.* An inspection shall be made to determine that packaging, packing, and marking requirements of section 5 are complied with. Defects shall be scored in accordance with table V. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

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TABLE V. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted, incorrect; illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as loose blocking or strapping and inadequate stapling.
	Bulging or distortion of container.

4.4 Tests.

4.4.1 *Hydrostatic strength.* Couplings shall be hydrostatically tested using water as the test fluid. This test may be made with hose attached in connection with test of the hose. For each one that fails in this test, two additional couplings shall be tested. Failure of any one of the additional couplings to pass the test shall cause rejection of the lot (see 3.5).

4.4.2 *Chemical analysis.* Chemical analysis shall be in accordance with method 111.1 or 112.1 of Fed. Test Method Std. No. 151. Two couplings selected from the samples in 4.2.3 shall be used for chemical analysis (see 3.4).

4.4.2.1 *Wet chemical analysis.* A composite sample shall be at least 2 ounces of chips or drillings. Loose ferritic metal shall be removed from the sample using a magnet. The sample shall also be free from dirt, oil, grit, and other foreign material.

4.4.2.2 *Spectrochemical analysis.* Specimens for spectrochemical analysis shall be prepared in accordance with method 112.1 of Fed. Test Method Std. No. 151.

4.5 Rejection. If any sample fails to conform to this specification, the entire lot shall be rejected, subject to retest provisions in Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Cleaning.* The couplings shall be cleaned by cleaning process C-3 in accordance with MIL-P-116.

5.1.1.2 *Drying.* The couplings shall be dried by drying procedure D-1 in accordance with MIL-P-116.

5.1.1.3 *Preservative application.*

5.1.1.3.1 *Types III and IV.* All unpainted exterior surfaces of the couplings subject to corrosion shall be coated with type P-7 preservative in accordance with MIL-P-116.

5.1.1.4 *Unit packaging.*

5.1.1.4.1 *Types I and II.* Couplings shall be packaged method III with all openings closed with material conforming to MIL-B-121, grade A, class 2 and secured with PPP-T-76 tape.

5.1.1.4.2 *Types III and IV.* Couplings shall be packaged method I with all openings and coated surfaces covered with material conforming to MIL-B-121, grade A, class 2, and secured with PPP-T-76 tape.

5.1.2 *Level C.* The couplings shall be packaged in accordance with commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Couplings of like description shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type or PPP-B-621, class 2. Blocking and bracing shall be provided to prevent movement. The box shall be closed and strapped in accordance with the appendix to the applicable box specification. The gross weight shall not exceed 200 pounds.

5.2.2 *Level B.* Couplings of like description shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type or PPP-B-621, class 1. Blocking and bracing shall be provided to prevent movement. The box shall be closed in accordance with the appendix to the applicable box specification. Unless otherwise specified, strapping shall be provided. The gross weight shall not exceed 150 pounds.

5.2.3 *Level C.* The cuplings shall be packed

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to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to markings required by the contract or order, the package shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Couplings covered by this specification are intended for use with hose manufactured for oil-suction and discharge.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Type and size of couplings (see 1.2).
(*c*) Thread symbol; NH or NPSH (see 3.1).
(*d*) Clamp for type III, if required (see 3.3).
(*e*) Companion flange, if required, types III and IV (see 3.3).
(*f*) Size of lot, if different from 4.2.1.
(*g*) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).

6.3 For civil agencies. When level B packaging is required for civil agency procurements, the requirements of 5.1.1 level A packaging shall apply.

MILITARY INTERESTS:

Military custodians:
Army—MO(ERDL)
Navy—YD
Air Force—69

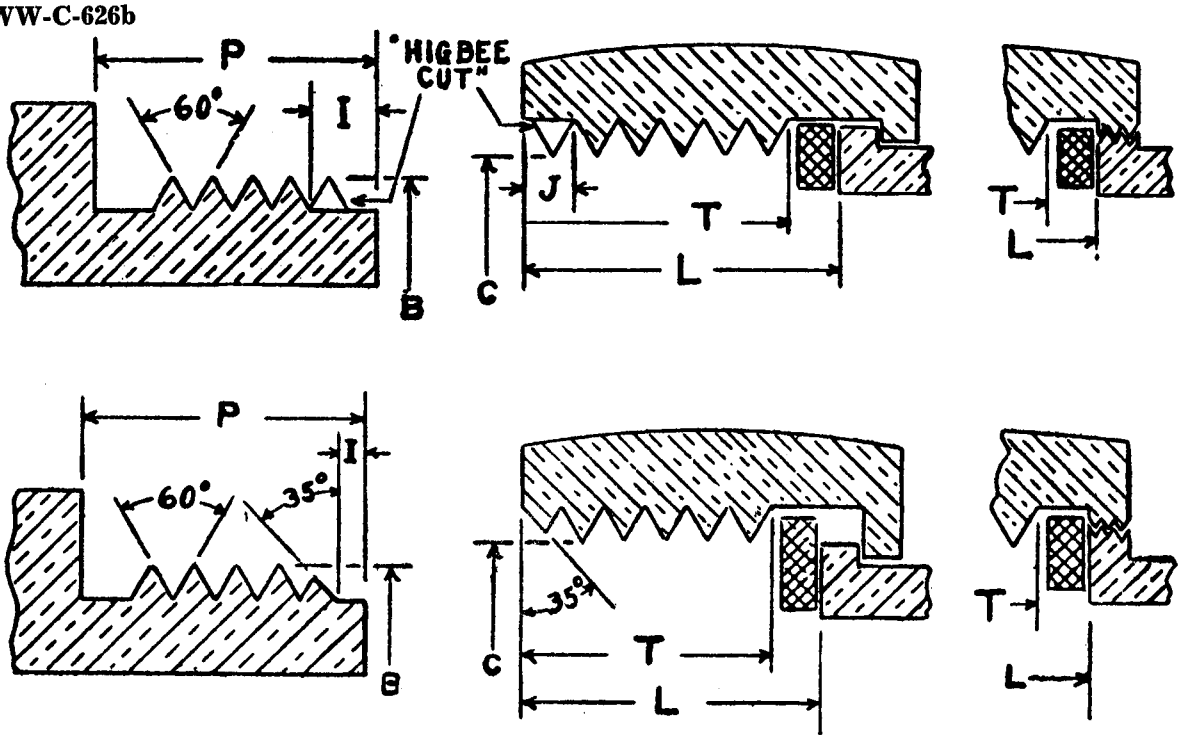
Review interests:
MO, YD, 69

User interests:
EL, MI, MU, GL

MILITARY COORDINATING ACTIVITY:
Army—MO(ERDL)

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Figures for hose coupling threads.

Table for hose coupling thread

Nominal size	Diameter B	I	J	L	P	T	Threads per inch	Full threads in T[1]
Inches 4	.....	5/16	1/4	1-1/16	1-1/8	13/16	6	4-1/4

[1] For all other sizes see table "Lengths of Thread for American National Hose-Coupling Threads" in NBS H-28.

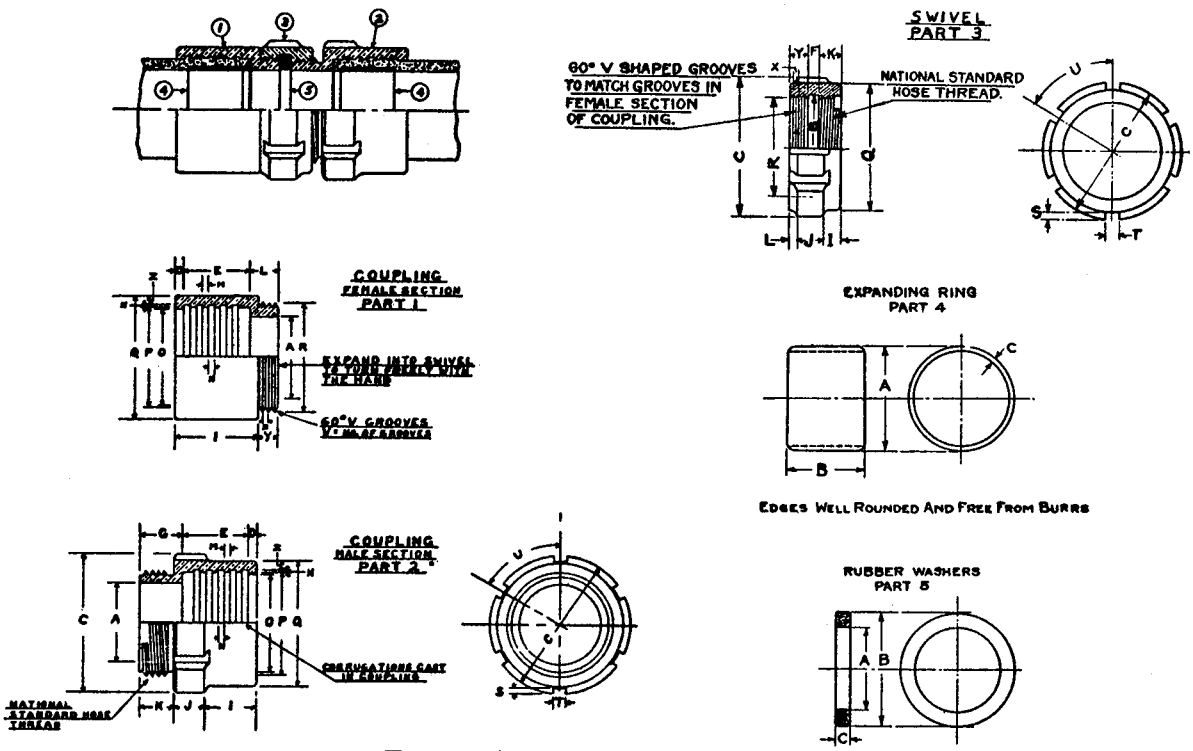
Table for coupling threads

Nominal size (inches)	Threads per inch	Pitch	Depth of thread	Pitch diameter			Minor diameter		
				Maximum	Tolerance	Minimum	Maximum	Tolerance	Minimum
		(Inch)	*(Inch)*	*(Inches)*	*(Inch)*	*(Inches)*	*(Inches)*	*(Inch)*	*(Inches)*
1-1/2	11-1/2	0.08696	0.05648	1.8408	0.0085	1.8323	1.7928	0.0170	1.7758
2	11-1/2	.08696	.05648	2.3148	.0085	2.3063	2.2668	.0170	2.2498
2-1/2	7-1/2	.13333	.08660	3.0130	.0160	2.9970	2.9424	.0320	2.9104
3	6	.16667	.10825	3.5486	.0180	3.5306	3.4583	.0360	3.4223
3-1/2	6	.16667	.10825	4.1736	.0180	4.1556	4.0833	.0360	4.0473
4	6	.16667	.10825	4.8380	.0180	4.820	4.7477	.0360	4.7117

Table for nipple threads

Nominal size (inches)	Threads per inch	Pitch	Depth of thread	Pitch diameter			Minor diameter		
				Maximum	Tolerance	Minimum	Maximum	Tolerance	Minimum
		(Inch)	*(Inch)*	*(Inches)*	*(Inch)*	*(Inches)*	*(Inches)*	*(Inch)*	*(Inches)*
1-1/2	11-1/2	0.08696	0.05648	1.8223	0.0085	1.8138	1.8788	0.0170	1.8618
2	11-1/2	.08696	.05648	2.2963	.0085	2.2878	2.3528	.0170	2.3358
2-1/2	7-1/2	.13333	.08660	2.29820	.0160	2.29660	3.0686	.0320	3.0366
3	6	.16667	.10825	3.5156	.0180	3.4976	3.6239	.0360	3.5879
3-1/2	6	.16667	.10825	4.1356	.0180	4.1176	4.2439	.0360	4.2079
4	6	.16667	.10825	4.7999	.0180	4.7819	4.9082	.0360	4.8722

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Figures for parts of type I, couplings.

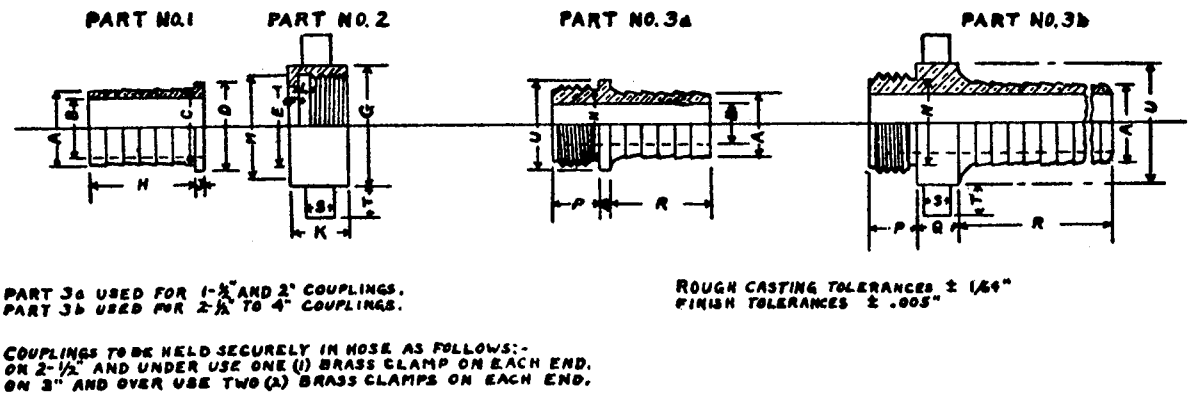
Table of dimensions for type I, couplings

Part No.	A	B	C	D	E	F	G	H (minimum)	I	J	K	L
1	4-1/64[1]	.....	6-3/16	1/8	2-5/8	.....	.....	1/32	3	.....	.....	1
2	4-1/64[1]	.....	6-3/16	1/8	2-5/8	.....	1-3/8	1/32	2-5/16	11/16	1-1/8	.....
3	.....	4-15/16	6-3/16	.....	.....	1/4	.....	.....	9/16	11/16	1-3/16	9/16
4	4	2-5/8	0.070 (min.) .094 (max.)	.....	.....	.....	.....	.....	.....	.....	.....	.....
5	4-1/16	4-15/16	3/16	.....	.....	.....	.....	.....	.....	.....	.....	.....

Part No.	M	N	O	P	Q	R	S	T	U	V	X	Y	Z (minimum)
1	3/64	9/64	5-3/16[2]	5-1/4[2]	5-3/4	4-3/4	.....	.....	.....	8	3/32	3/4	3/16
2	3/64	9/64	5-3/16[2]	5-1/4[2]	5-3/4	.....	3/16	1/2	45°	..	.....	.....	3/16
3	.....	.....	.....	.....	5-3/4	4-3/4	3/16	1/2	45°	8	3/32	3/4	.....

[1] Minimum + 1/64 inch maximum.
[2] ± 1/8 inch to suit hose.

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Figures for parts of type II, couplings.

Table of dimensions for type II, couplings

Size	A	B	C	D	E	F	G	H	J	K	L
Inches											
1-1/2	1-9/16	1-5/16	1-19/32	1-47/64	1-39/64	1/8	2-1/8	1-3/4	1/8	27/32	1/4
2	2-1/16	1-13/16	2-3/32	2-13/64	2-7/64	5/32	2-5/8	2-3/16	5/32	1-1/32	9/32
2-1/2	2-9/16	2-5/16	2-5/8	2-13/16	2-41/64	3/16	3-11/32	2-1/2	5/32	1-9/32	1-3/32
3	3-1/16	2-3/4	3-9/64	3-3/8	3-5/32	3/16	3-15/16	3-1/4	5/32	1-13/32	1-3/32
3-1/2	3-9/16	3-1/4	3-41/64	3-15/16	3-21/32	3/16	4-5/8	3-7/8	1/4	1-1/2	1/2
4	4-1/16	3-11/16	4-9/64	4-5/8	4-5/32	3/16	5-5/16	4-5/8	1/4	1-1/2	1/2

Size	M[1]	N[2]	P	Q	R	S	T	U	V	W	X (minimum)[3]
Inches											
1-1/2	.....	.....	5/8	1/8	1-5/8	15/32	7/16	2-1/8	1-7/8	1-3/8	1/8
2	.....	.....	3/4	3/16	1-15/16	1/2	1/2	2-5/8	2-11/32	1-7/8	1/8
2-1/2	.....	.....	1	3/4	2-3/8	1/2	1/2	3-11/32	3-1/16	2-3/8	3/16
3	.....	.....	1-1/8	3/4	3-1/4	9/16	5/8	3-15/16	3-5/8	2-7/8	3/16
3-1/2			1-1/8	3/4	3-9/16	9/16	5/8	4-5/8	4-1/4	3-3/8	3/16
4	.....	.....	1-1/8	3/4	4-9/16	3/4	3/4	5-5/16	4-15/16	3-13/16	3/16

[1] To outside diameter of thread.
[2] Relieve at least to bottom of thread.
[3] All + 1/32 inch maximum.

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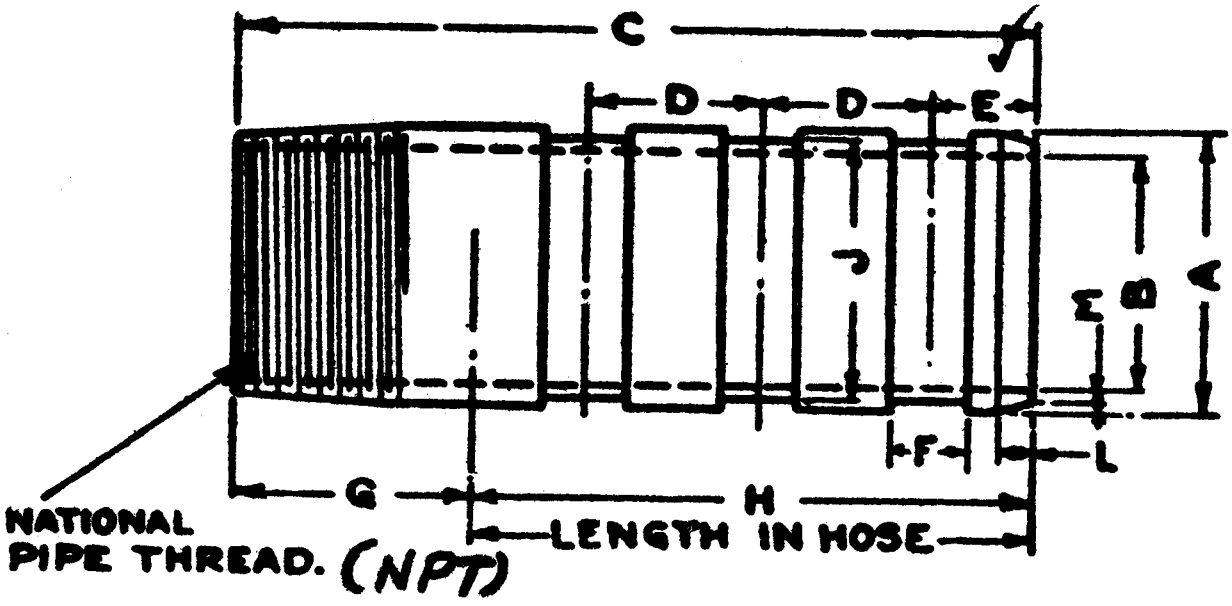


Figure for type III, coupling (nipple).

Table of dimensions for type III, couplings

Size	A	B	C	D	E	F	G	H	J	L	No. of groove	M
Inches												
2	2.37	2.06	7	2-1/8	1-1/16	1-1/8	2-1/2	4-1/2	2-1/4	1/4	2	1/16
2-1/2	2.87	2.46	8	2-1/2	1-1/4	1-1/4	3	5	2-3/4	1/4	2	1/16
3	3.50	3.06	8	2-1/2	1-1/4	1-1/4	3	5	3-3/8	1/4	2	1/16
3-1/2	4.00	3.54	8	2-1/2	1-1/4	1-1/4	3	5	3-7/8	1/4	2	1/16
4	4.50	4.02	10	2-3/4	1-5/8	1-1/4	4	6	4.37	1/2	2	1/16
5	5.56	5.05	11	3	1-7/8	1-1/4	4	7	5.37	1/2	2	1/16
6	6.62	6.06	16-1/2	3-1/2	2-1/4	1-1/2	5	11-1/2	6.43	1/2	3	1/16
8	8.62	8.07	17-1/2	3-1/2	2-1/4	1-1/2	6	11-1/2	8.43	1/2	3	1/8
10	10.75	10.19	20	4	2-3/4	1-1/2	7	13	10.5	1/2	3	1/8
12	12.75	12.09	20	4	2-3/4	1-1/2	7	13	12.5	3/4	3	1/8

All dimensions are in inches.
All burrs and sharp projections to be removed.
Standard wrought iron and steel pipe nipples for oil-suction and discharge hose.

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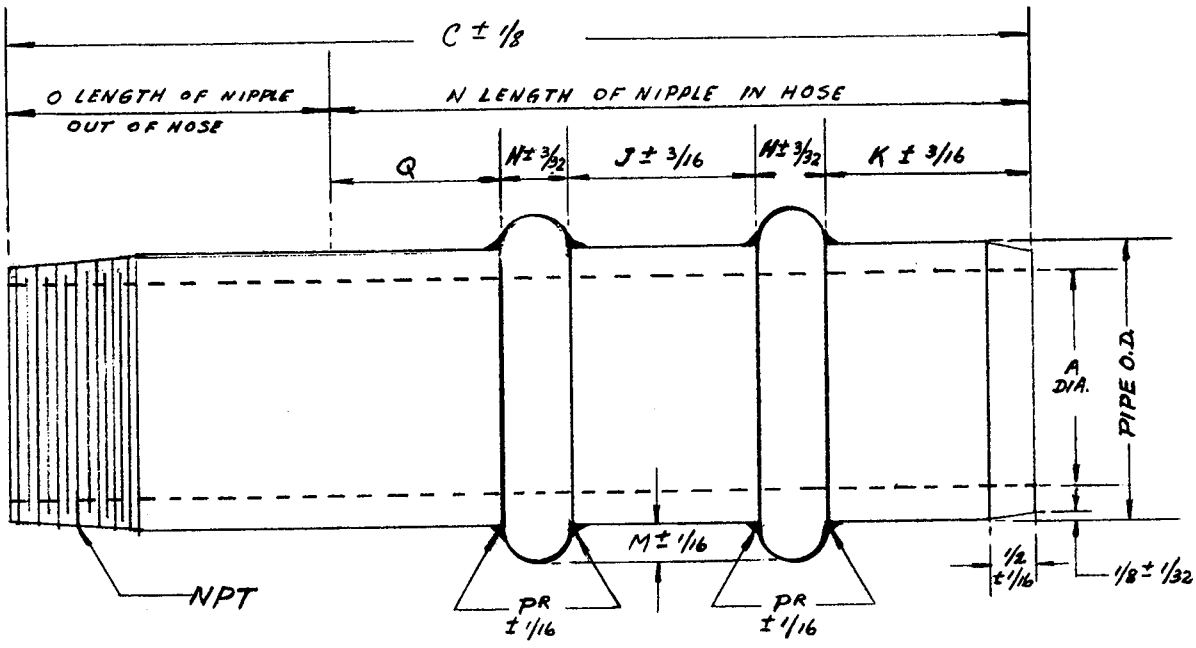


Figure for type IV, coupling (nipple).

Table of dimensions for type IV, coupling (nipple)

Size	Pipe O.D.	A[1]	C	H	J	K	M	N	O	P	Q
Inches											
2	2.37	2.03	12-1/2	3/4	3	3	3/8	9-1/2	3	3/16	2
2-1/2	2.87	2.53	13-1/2	3/4	3-1/2	3-1/2	3/8	10-1/2	3	3/16	2
3	3.50	3.03	15-1/2	3/4	4	3-1/2	3/8	11-1/2	4	3/16	2-1/2
3-1/2	4.00	3.53	15-1/2	3/4	4	3-1/2	3/8	11-1/2	4	3/16	2-1/2
4	4.50	4.03	16	3/4	4	4	3/8	12	4	3/16	2-1/2
5	5.56	5.03	18-1/2	1	4-1/2	4-1/2	1/2	13-1/2	5	3/16	2-1/2
6	6.62	6.03	21	1	5	5	1/2	15	6	3/16	3
8	8.62	8.03	21	1	5	5	1/2	15	6	3/16	3
10	10.75	10.03	23-1/2	1-1/4	6	5	5/8	16-1/2	7	1/4	3
12	12.75	12.03	24	1-1/2	6	5	3/4	17	7	5/16	3

[1] Dimension is minimum.
All dimensions are in inches.

☆ U. S. GOVERNMENT PRINTING OFFICE: 1966—253348/97

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WW-C-00626a (GSA-FSS)
May 14, 1964
INTERIM REVISION OF
Fed. Spec. WW-C-626
February 7, 1933

INTERIM FEDERAL SPECIFICATION

COUPLING, HOSE (OIL-SUCTION AND DISCHARGE)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D. C. 20407, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized Federal agencies to use this Interim Federal Specification as a valid exception to Federal Specification WW-C-626 dated February 7, 1933.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers four types of hose couplings for oil suction and discharge.

1.2 Classification. Couplings shall be of the following types and sizes as specified (see 6.2):

1.2.1 Types:

Type I - Slotted expansion
Type II - Lug, ribbed shank
Type III - Iron pipe nipples
Type IV - Iron pipe nipples (banded)

1.2.2 Size:

Type I - 4 inches only
Type II - 1-1/2, 2, 2-1/2, 3 and 4 inches
Type III and IV - 2, 2-1/2, 3, 4, 5, 6, 8, 10 and 12 inches

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS.

2.1 Specifications and Standards. The following specifications and standards of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-B-621 - Brass; Castings, Leaded Yellow.
QQ-B-626 - Brass, Leaded and Non-leaded: Rod, Shapes, Forgings and Flat Products, With Finished Edges (Bar, Flat Wire and Strip).
WW-P-406 - Pipe, Steel (Seamless and Welded) (For Ordinary Use).
WW-P-441 - Pipe; Wrought Iron (Welded, Black or Zinc Coated).
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 4730

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Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue if effect on date of invitation for bids shall apply:

National Bureau of Standards
Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies of these screw-thread standards should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402).

3. REQUIREMENTS.

3.1 Design. Couplings shall conform to the respective figures and tables incorporated herein. Threads shall conform to Sections VII and X of H28 Part II.

3.2 Finish. All non-mating surfaces shall be finished reasonably smooth by grinding, polishing, shot blasting, or sand blasting, and shall be free from burrs. All mating and gasket surfaces shall be 125 rms finish or better and all machine tolerances, plus or minus 0.010 inch. The corrugations of both male and female sections of coupling shall be left rough so as to bind the hose.

3.3 Parts. A complete set of Types I, II, III and IV couplings shall consist of the following parts:

Type I:
Part 1. Female section.
Part 2. Male section.
Part 3. Swivel of female section.
Part 4. Expansion rings, one each for male and female sections.
Part 5. Rubber washer, fitting recess in swivel.

Type II:
Part 1. Female section.
Part 2. Male section.
Part 3. Swivel of female section.
Part 4. Clamps for attaching hose.
Part 5. Washer, fitting recess in swivel.

Type III:
Part 1. Nipple.
Part 2. Clamp, if specified. (see 6.2)
Part 3. Standard companion flange, if specified. (see 6.2)

Type IV:
Part 1. Nipple.
Part 2. Clamp, if specified. (see 6.2)
Part 3. Standard companion flange, if specified. (see 6.2)

3.4 Materials.

3.4.1 Type I and II. All metal parts except expansion rings shall be drop-forged brass conforming to QQ-B-626, Composition 21, cast yellow brass conforming to Composition C of QQ-B-621, or of a yellow brass conforming to chemical and physical properties in Tables I and II.

TABLE I - Chemical Composition

	Minimum	Maximum
Copper, percent	82.00	84.00
Tin, percent	2.5	3.5
Lead, percent	2.5	3.5
Zinc, percent	9.0	11.0
Phosphorus, percent	- - -	0.03
Antimony, percent	- - -	0.25
Nickel, percent	0.25	0.50
Other elements, other than copper, tin, lead, zinc and nickel, percent 1/	- - -	0.50

1/ Includes iron maximum percent 0.35.

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TABLE II - Physical Properties

	Minimum	Maximum
Tensile strength, lb/in 2/	31,000	- - - - - - - - -
Elongation in 2 inch, percent	25	- - - - - - - - -
Reduction in area, percent	20	30
Brinell hardness (500 kg. for 30 seconds)	50	60

3.4.2 Type I, part 4 expansion rings. Expansion rings used in Type I, coupling shall be made from seamless drawn brass tubing, conforming to either of the compositions in Table III.

TABLE III - Material Composition

Copper	Tin (maximum)	Zinc	Lead (maximum)	Iron (maximum)
Per cent	Per cent	Per cent	Per cent	Per cent
79-86	- - - - - - - - -	Remainder	0.20	0.05
59-68	1.5	Remainder	3.50	.20

3.4.3 Types III and IV. Nipples shall be of steel pipe conforming to WW-P-406, Weight A, or Wrought Iron Pipe conforming to WW-P-441, Class A. Bands required for type IV coupling shall be of steel and welded all around the diameter of coupling. Clamps when required shall be of brass having a copper content of not less than 68 per cent copper.

3.5 Hydrostatic pressure. The couplings shall be capable of withstanding the respective hydrostatic pressure for the sizes listed below:

Size inches	Pressure pounds
4	250
6	200
8	150

The pressures for other sizes shall be the same as required for the hose.

3.6 Workmanship. Couplings shall be free of burrs, cracks, sharp edges, pits or any other defects which would affect their serviceability for their intended purpose.

4. SAMPLING, INSPECTION AND TEST PROCEDURE.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. The sample size levels of inspection shall be in accordance with Military Standard MIL-STD-105.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all couplings of the same size and type submitted for inspection at the same time.

4.2.2 Visual and dimensional examination of end item. Sampling of couplings for these characteristics shall be in accordance with inspection level S-4.

4.2.3 Samples for test. Samples shall be selected as follows; samples for test may be obtained from those selected in 4.2.3.

Sample Size	Level or Number
1-25	0
26-50	S-1
51 and above	S-2

4.3 Inspection.

4.3.1 Examination of end item. The samples selected in 4.2.2 shall be examined for defects as listed in Table IV. The AQL's shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

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TABLE IV - Classification of defects, end item

Defects	Major	Minor
Cracks	X	
Porosity visible on surface		X
Metal protruding on parting or flash line of cast or forged units	X	
Threads rough or eccentric	X	
Sharp burrs or slivers	X	
Parts missing or wrong size	X	
Assembly damaged or defective not affecting function or serviceability		X

4.3.1.1 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with. Defects shall be scored in accordance with Table V. For examination of contents the sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

TABLE V - Classification of preparation for delivery defects

Examine	
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as loose blocking or strapping and inadequate stapling. Bulging or distortion of container.

4.3.2 Lot acceptance tests. Samples selected in 4.2.3 shall be subjected to the tests in 4.4, AQL 10 percent defective.

4.4 Tests.

4.4.1 Hydrostatic strength. Couplings shall be hydrostatically tested using water as the test fluid. This test may be made with hose attached in connection with test of the hose. For each one that fails in this test, two additional couplings shall be tested. Failure of any one of the additional couplings to pass the test shall cause rejection of the lot (see 3.5).

4.4.2 Chemical analysis. Chemical analysis. Chemical analysis shall be in accordance with method 111.1 or 112.1 of Federal Test Method Standard No. 151. Two couplings selected from the samples in 4.2.4 shall be used for chemical analysis (see 3.4).

4.4.2.1 Wet chemical analysis. A composite sample shall be at least 2 ounces of chips or drillings. Loose ferritic metal shall be removed from the sample using a magnet. The sample shall also be free from dirt, oil, grit, and other foreign material.

4.4.2.2 Spectrochemical analysis. Specimens for spectrochemical analysis shall be prepared in accordance with method 112.1 of Federal Test Method Standard No. 151.

4.5 Rejection. If any sample fails to conform to this specification, the entire lot shall be rejected, subject to retest provisions in Federal Test Method Standard No. 151.

5. PREPARATION FOR DELIVERY.

(For Civil agency procurement the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102).

5.1 Preservation and packaging. Preservation and packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A or B.

5.1.1.1 Cleaning. The couplings shall be cleaned by cleaning process C-3 in accordance with MIL-P-116.

5.1.1.2 Drying. The couplings shall be dried by drying procedure D-1 in accordance with MIL-P-116.

5.1.1.3 Preservative application.

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5.1.1.3.1 Type III and type IV. All unpainted exterior surfaces of the couplings subject to corrosion shall be coated with type P-7 preservative in accordance with MIL-P-116.

5.1.1.4 Unit packaging.

5.1.1.4.1 Types I and II. Couplings shall be packaged method III with all openings closed with material conforming to MIL-B-121, grade A, class 2 and secured with PPP-T-76 tape.

5.1.1.4.2 Types III and IV. Couplings shall be packaged method I with all openings and coated surfaces covered with material conforming to MIL-B-121, grade A, class 2 and secured with PPP-T-76 tape.

5.1.2 Level C. The couplings shall be packaged in accordance with commercial practice.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Couplings of like description packaged as specified in 5.1.1.4 shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type or PPP-B-621, class 2. Blocking and bracing shall be provided to prevent movement. The box shall be closed and strapped in accordance to the appendix to the applicable box specification. The gross weight shall not exceed 200 pounds.

5.2.2 Level B. Couplings of like description packaged as specified in 5.1.1.4 shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type of PPP-B-621, class 1. Blocking and bracing shall be provided to prevent movement. The box shall be closed in accordance with the appendix to the applicable box specification. Unless otherwise specified, strapping shall be provided. The gross weight shall not exceed 150 pounds.

5.2.3 Level C. The couplings shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the packaged and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. Couplings covered by this specification are intended for use with hose manufactured for oil-suction and discharge.

6.2 Ordering data. Purchasers should exercise sny desired options offered herein, and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and size of couplings (see 1.2).
(c) Thread symbol; NH or NPSH (see 3.1).
(d)Clamp for types III and IV, if required (see 3.3).
(e) Companion flange, if required types III and IV (see 3.3).
(f) Size of lot, if different from 4.2.1.
(g) Levels of packaging and packing (see 5.1 and 5.2).

6.3 Transportation description. The transportation descriptions and minimum weights applicable to this commodity are: (to be furnished)

RAIL:

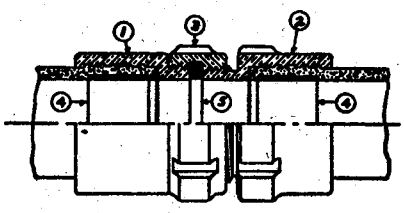
MOTOR:

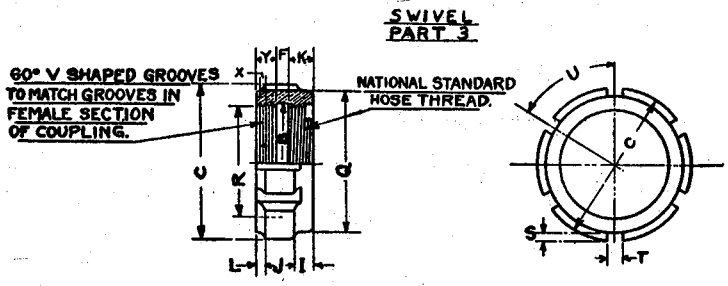
MILITARY INTERESTS:
Army - MO
Navy - SHIPS
Air Force - MOMS

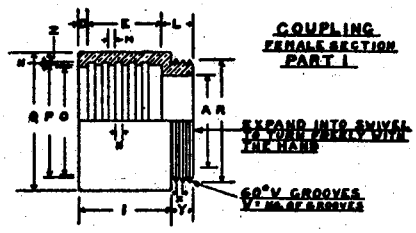
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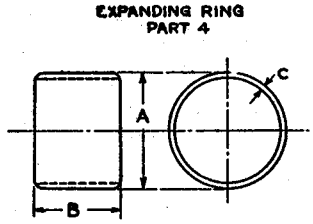
Figures for parts of Type I, Couplings

Coupling Assembly Type I

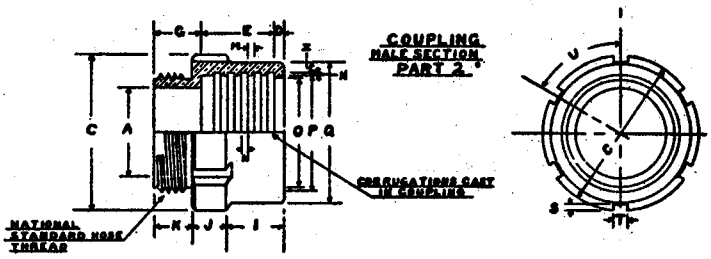








EDGES WELL ROUNDED AND FREE FROM BURRS



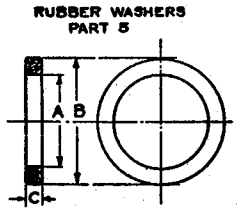


Table of dimensions for Type I, Couplings

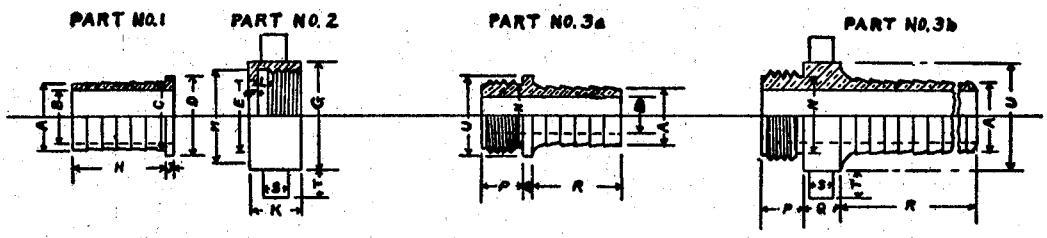
Part No.	A	B	C	D	E	F	G	H(minimum)	I	J	K	L
1 ---	4-1/64 1/	-------	6-3/16	1/8	2-5/8	---	-----	1/32	3	-----	-----	1
2 ---	4-1/64 1/	-------	6-3/16	1/8	2-5/8	---	1-3/8	1/32	2-5/16	11/16	1-1/8	----
3 ---	-------	4-15/16	6-3/16	---	-----	1/4	-----	----	9/16	11/16	1-3/16	9/16
4 ---	4	2-5/8	.070(min.) .094(max.)	---	-----	---	-----	----	-------	-----	-------	----
5 ---	4-1/16	4-15/16	3/16	---	-----	---	-----	----	-------	-----	-------	----

Part No.	M	N	O	P	Q	R	S	T	U	V	X	Y	Z(minimum)
1 ---	3/64	9/64	5-3/16 2/	5-1/4 2/	5-3/4	4-3/4	----	---	---	8	3/32	3/4	3/16
2 ---	3/64	9/64	5-3/16 2/	5-1/4 2/	5-3/4	-----	3/16	1/2	45°	-	----	---	3/16
3 ---	----	----	--------	-------	5-3/4	4-3/4	3/16	1/2	45°	8	3/32	3/4	----

1/ Minimum ± 1/64 inch maximum.
2/ ± 1/8 inch to suit hose.

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Figures for parts of Type II, Couplings



PART 3a USED FOR 1-½" AND 2" COUPLINGS.
PART 3b USED FOR 2-½" TO 4" COUPLINGS.

ROUGH CASTING TOLERANCES ± 1/64"
FINISH TOLERANCES ± .005"

COUPLINGS TO BE HELD SECURELY IN HOSE AS FOLLOWS:-
ON 2-½" AND UNDER USE ONE (1) BRASS CLAMP ON EACH END.
ON 3" AND OVER USE TWO (2) BRASS CLAMPS ON EACH END.

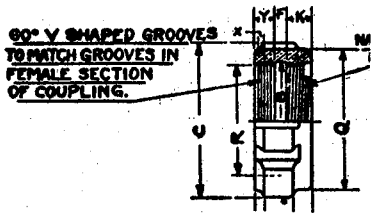


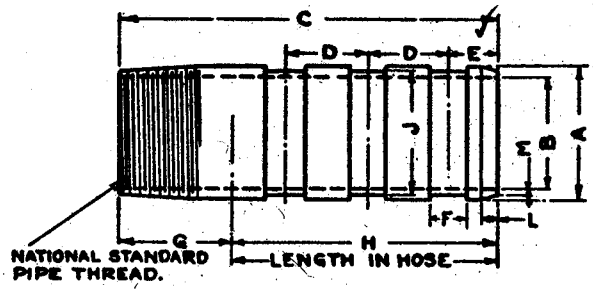
Table of dimensions for Type II, Couplings

Size	A	B	C	D	E	F	G	H	J	K	L
inches											
1-1/2	1-9/16	1-5/16	1-19/32	1-47/64	1-39/64	1/8	2-1/8	1-3/4	1/8	27/32	1/4
2	2-1/16	1-13/16	2-3/32	2-13/64	2-7/64	5/32	2-5/8	2-3/16	5/32	1-1/32	9/32
2-1/2	2-9/16	2-5/16	2-5/8	2-13/16	2-41/64	3/16	3-11/32	2-1/2	5/32	1-9/32	1-3/32
3	3-1/16	2-3/4	3-9/64	3-3/8	3-5/32	3/16	3-15/16	3-1/4	5/32	1-13/32	1-3/32
3-1/2	3-9/16	3-1/4	3-41/64	3-15/16	3-21/32	3/16	4-5/8	3-7/8	1/4	1-1/2	1/2
4	4-1/16	3-11/16	4-9/64	4-5/8	4-5/32	3/16	5-5/16	4-5/8	1/4	1-1/2	1/2

Size	M 1/	N 2/	P	Q	R	S	T	U	V	W	X(minimum) 3/
inches											
1-1/2	----	----	5/8	1/8	1-5/8	15/32	7/16	2-1/8	1-7/8	1-3/8	1/8
2	----	----	3/4	3/16	1-15/16	1/2	1/2	2-5/8	2-11/32	1-7/8	1/8
2-1/2	----	----	1	3/4	2-3/8	1/2	1/2	3-11/32	3-1/16	2-3/8	3/16
3	----	----	1-1/8	3/4	3-1/4	9/16	5/8	3-15/16	3-5/8	2-7/8	3/16
3-1/2			1-1/8	3/4	3-9/16	9/16	5/8	4-5/8	4-1/4	3-3/8	3/16
4	----	----	1-1/8	3/4	4-9/16	3/4	3/4	5-5/16	4-15/16	3-13/16	3/16

1/ To outside diameter of thread.
2/ Relieve at least to bottom of thread.
3/ All ± 1/32 inch maximum.

Figure for Type II, Coupling (Nipple)



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Table of dimensions for Type III, Couplings

Size	A	B	C	D	E	F	G	H	J	L	No. of groove	M
inches												
2	2.37	2.06	7	2-1/8	1-1/16	1-1/8	2-1/2	4-1/2	2-1/4	1/4	2	1/16
2-1/2	2.87	2.46	8	2-1/2	1-1/4	1-1/4	3	5	2-3/4	1/4	2	1/16
3	3.50	3.06	8	2-1/2	1-1/4	1-1/4	3	5	3-3/8	1/4	2	1/16
3-1/2	4.00	3.54	8	2-1/2	1-1/4	1-1/4	3	5	3-7/8	1/4	2	1/16
4	4.50	4.02	10	2-3/4	1-5/8	1-1/4	4	6	4.37	1/2	2	1/16
5	5.563	5.05	11	3	1-7/8	1-1/4	4	7	5.37	1/2	2	1/16
6	6.62	6.06	16-1/2	3-1/2	2-1/4	1-1/2	5	11-1/2	6.43	1/2	3	1/16
8	8.62	8.07	17-1/2	3-1/2	2-1/4	1-1/2	6	11-1/2	8.43	1/2	3	1/8
10	10.75	10.19	20	4	2-3/4	1-1/2	7	13	10.5	1/2	3	1/8
12	12.75	12.09	20	4	2-3/4	1-1/2	7	13	12.5	3/4	3	1/8

All dimensions are in inches.
All burrs and sharp projections to be removed.
Standard wrought iron and steel pipe nipples for oil suction and discharge hose.

Figure for Type IV, Coupling (Nipple)

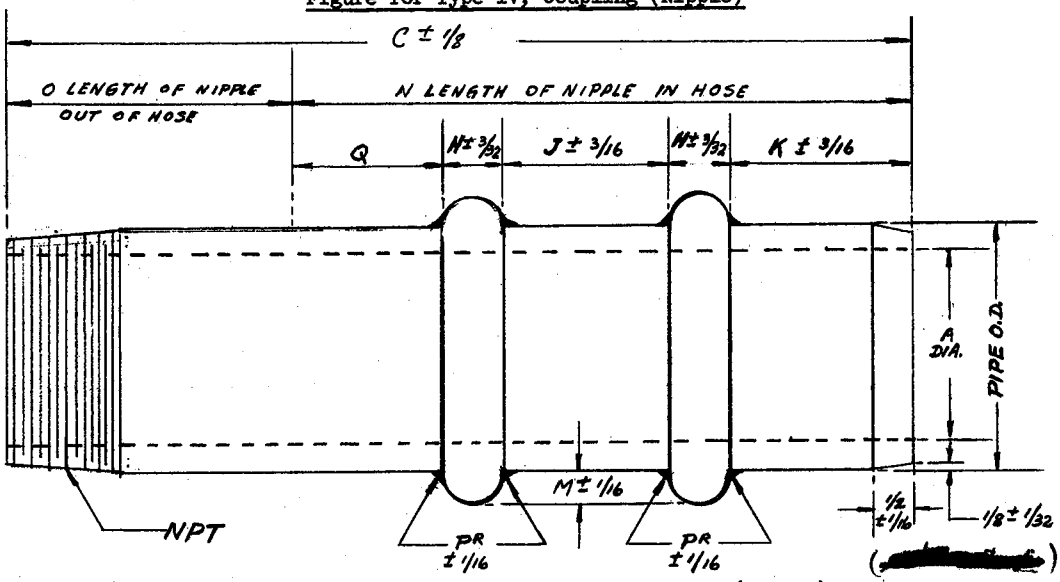


Table of dimensions for Type IV, Coupling (Nipple)

Size	Pipe O.D.	A 1/	C	H	J	K	M	N	O	P	Q
inches											
2	2.37	2.03	12-1/2	3/4	3	3	3/8	9-1/2	3	3/16	2
2-1/2	2.87	2.53	13-1/2	3/4	3-1/2	3-1/2	3/8	10-1/2	3	3/16	2
3	3.50	3.03	15-1/2	3/4	4	3-1/2	3/8	11-1/2	4	3/16	2-1/2
3-1/2	4.00	3.53	15-1/2	3/4	4	3-1/2	3/8	11-1/2	4	3/16	2-1/2
4	4.50	4.03	16	3/4	4	4	3/8	12	4	3/16	2-1/2
5	5.56	5.03	18-1/2	1	4-1/2	4-1/2	1/2	13-1/2	5	3/16	2-1/2
6	6.62	6.03	21	1	5	5	1/2	15	6	3/16	3
8	8.62	8.03	21	1	5	5	1/2	15	6	3/16	3
10	10.75	10.03	23-1/2	1-1/4	6	5	5/8	16-1/2	7	1/4	3
12	12.75	12.03	24	1-1/2	6	5	3/4	17	7	5/16	3

1/ Dimension is minimum.
All dimensions are in inches.

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Figures for Hose coupling threads

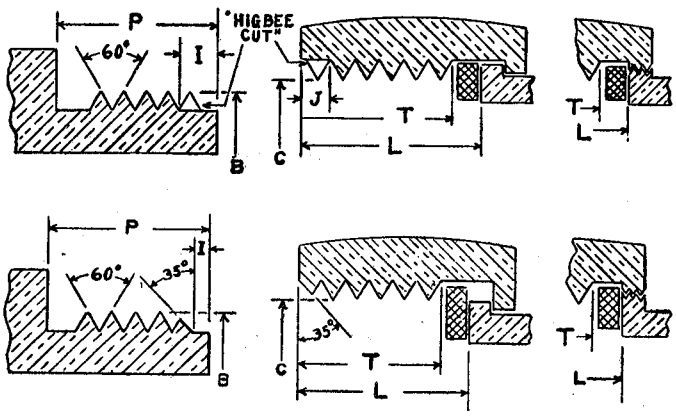


Table for hose coupling thread

Nominal size	Diameter B	I	J	L	P	T	Threads per inch	Full threads in T 1/
inches 4	---------	5/16	1/4	1-1/16	1-1/8	1-3/16	6	4-1/4

1/ For all other sizes see table "Lengths of Thread for American National hose-coupling threads" in H-28.

Table for coupling threads

Nominal size (inches)	Threads per inch	Pitch	Depth of thread	Pitch diameter			Minor diameter		
				Maximum	Tolerance	Minimum	Maximum	Tolerance	Minimum
		(inch)	(inch)	(inches)	(inch)	(inches)	(inches)	(inch)	(inches)
1-1/2	11-1/2	0.08696	0.05648	1.8408	0.0085	1.8323	1.7928	0.0170	1.7758
2	11-1/2	0.08696	0.05648	2.3148	0.0085	2.3063	2.2668	0.0170	2.2498
2-1/2	7-1/2	0.13333	0.08660	3.0130	0.0160	2.9970	2.9424	0.0320	2.9104
3	6	0.16667	0.10825	3.5486	0.0180	3.5306	3.4583	0.0360	3.4223
3-1/2	6	0.16667	0.10825	4.1736	0.0180	4.1556	4.0833	0.0360	4.0473
4	6	0.16667	0.10825	4.8380	0.0180	4.820	4.7477	0.0360	4.7117

Table for nipple threads

Nominal size (inches)	Threads per inch	Pitch	Depth of thread	Pitch diameter			Minor diameter		
				Maximum	Tolerance	Minimum	Maximum	Tolerance	Minimum
		(inch)	(inch)	(inches)	(inch)	(inches)	(inches)	(inch)	(inches)
1-1/2	11-1/2	0.08696	0.05648	1.8223	0.0085	1.8138	1.8788	0.0170	1.8618
2	11-1/2	0.08696	0.05648	2.2963	0.0085	2.2878	2.3528	0.0170	2.3358
2-1/2	7-1/2	0.13333	0.08660	2.29820	0.0160	2.29660	3.0686	0.0320	3.0366
3	6	0.16667	0.10825	3.5156	0.0180	3.4976	3.6239	0.0360	3.5879
3-1/2	6	0.16667	0.10825	4.1356	0.0180	4.1176	4.2439	0.0360	4.2079
4	6	0.16667	0.10825	4.7999	0.0180	4.7819	4.9082	0.0360	4.8722

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